

GCP Cloud HA VPN ↔ FortiGate Dual ISP BGP Integration

Last edited by [Smrutikanta Biswal](#) 1 hour ago

GCP Cloud HA VPN FortiGate Dual-ISP BGP Integration

Project Overview

This project established a highly available site-to-site VPN connection between an on-premises **FortiGate firewall running FortiOS 7.4.11** and a **Google Cloud VPC** using **Cloud HA VPN and BGP**. The design uses two independent ISP paths on the FortiGate, with each ISP terminating a separate IPsec tunnel to the GCP HA VPN gateway.

The integration is fully operational and has been validated in production, including IPsec tunnel establishment, dual BGP sessions, bidirectional route exchange, and ISP failover behavior.

Why BGP Over Static Routing?

BGP was selected instead of static routing to provide dynamic route exchange and better resilience across the two VPN paths. With BGP, the FortiGate and GCP Cloud Router dynamically learn the required network prefixes and can react to path availability without maintaining separate static routes for every remote network. This makes the design easier to operate and provides a cleaner foundation for redundant VPN connectivity and ISP failover. Route advertisements are also explicitly scoped so that only the intended LAN prefixes are propagated, avoiding accidental advertisement of WAN, transit, or other infrastructure networks.

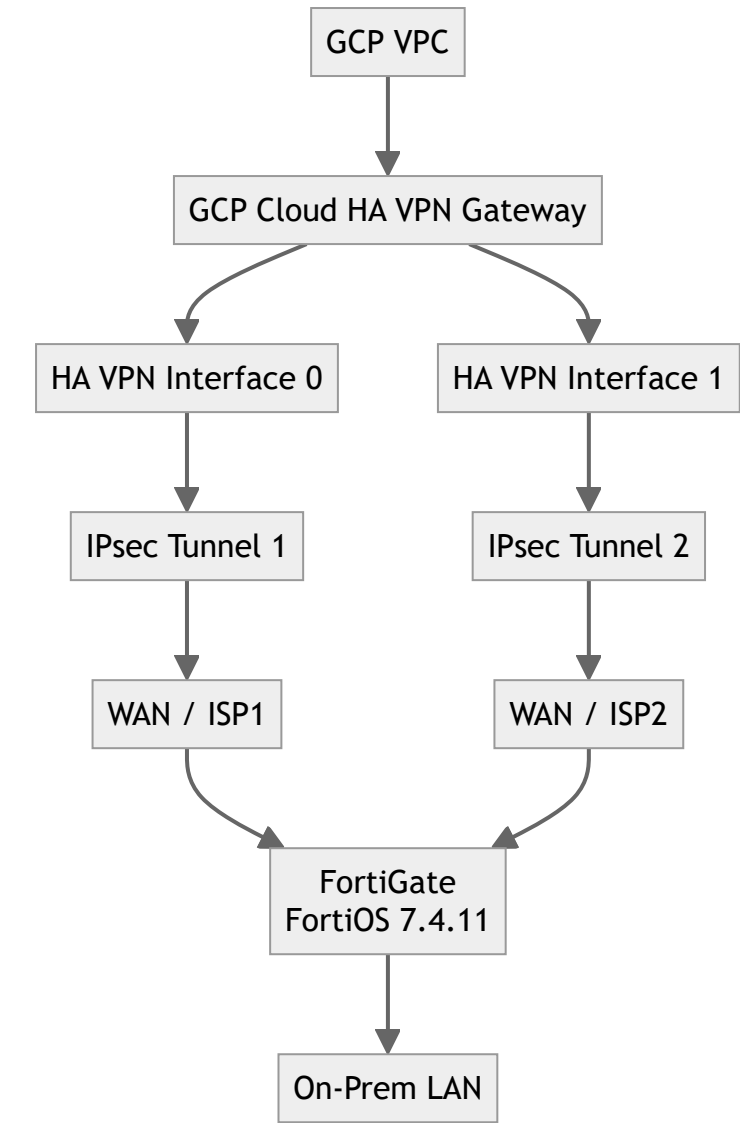
Table of Contents

- [Project Overview](#)
- [Why BGP Over Static Routing?](#)
- [Architecture](#)
 - [High-Level Traffic Flow](#)
 - [BGP Session View](#)
- [Components](#)
- [Setup Steps](#)
 - [1. GCP Configuration](#)
 - [2. FortiGate IPsec Configuration](#)
 - [3. FortiGate BGP Configuration](#)
 - [4. Firewall Policies](#)
- [Verification & Testing](#)
- [Lessons Learned / Gotchas](#)
- [Production Outcome](#)

Architecture

High-Level Traffic Flow

The architecture consists of a GCP VPC connected to a Cloud HA VPN gateway. The HA VPN gateway provides two VPN interfaces, with each interface mapped to a separate FortiGate ISP/public-IP path.



Traffic and Redundancy Model

The two VPN paths are independent:

Path	GCP Side	On-Premises Side	Routing
Path 1	HA VPN Interface 0	ISP1 / WAN1	BGP
Path 2	HA VPN Interface 1	ISP2 / WAN2	BGP

Each path consists of:

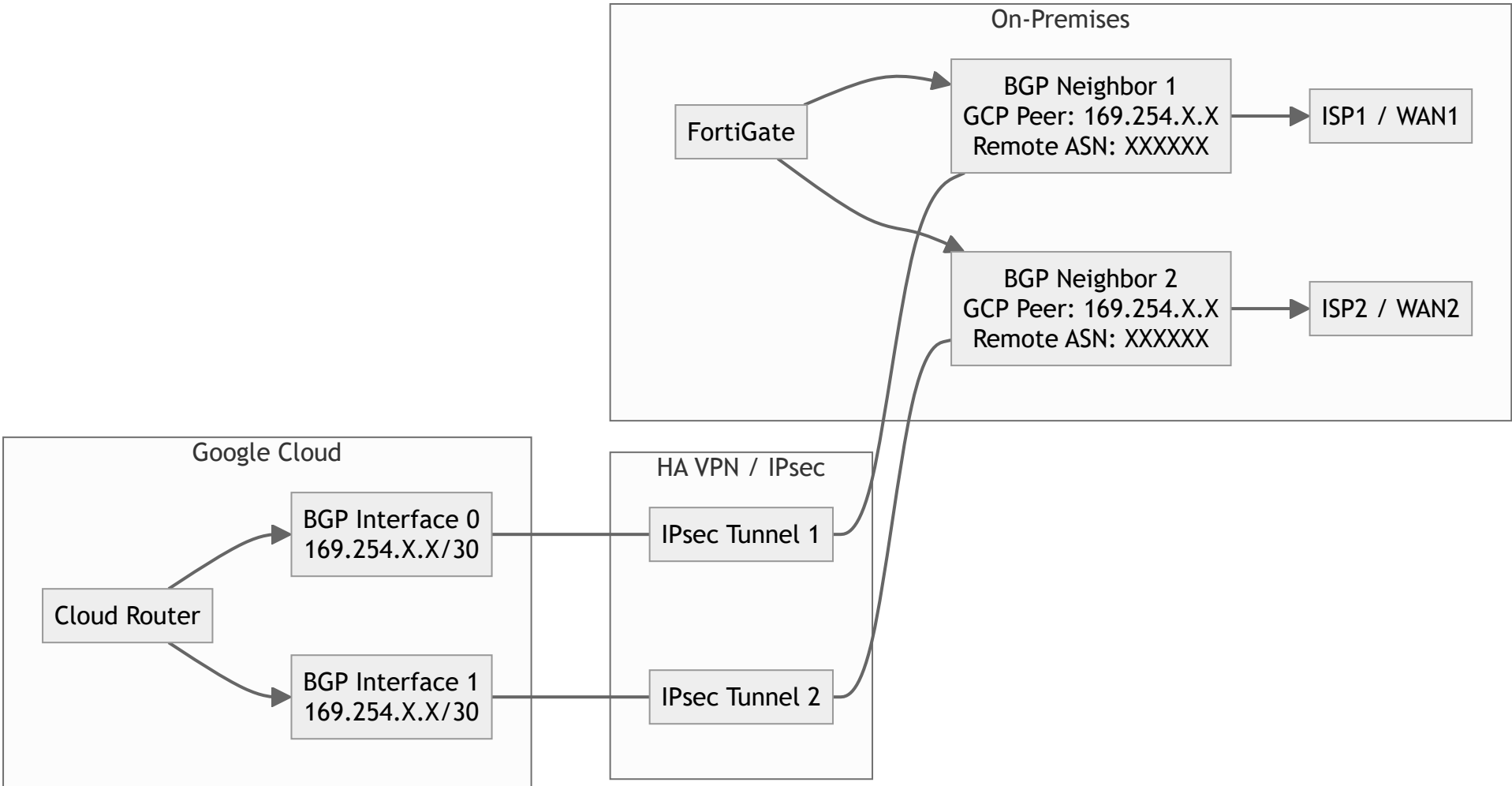
GCP VPC → HA VPN interface → IPsec tunnel → FortiGate WAN/ISP → FortiGate routing → LAN

The two paths operate as redundant VPN connections, with BGP providing the dynamic routing control plane.

BGP Session View

The GCP Cloud Router provides the BGP control plane for the VPN connection. Each HA VPN tunnel has its own BGP adjacency with the corresponding FortiGate BGP neighbor.

The tunnel interfaces use the **GCP-assigned link-local /30 addressing**. The actual addresses are environment-specific and are represented below using placeholders.



BGP Relationship

Each VPN path has a dedicated BGP session:



The FortiGate uses two static BGP neighbor definitions, one for each GCP Cloud Router BGP peer.

The BGP tunnel addresses are based on GCP's automatically assigned **169.254.X.X/30 link-local ranges**.

Components

Component	Role
GCP VPC	Cloud-side network hosting the required workloads/subnets
Cloud HA VPN Gateway	Highly available GCP VPN termination point
Cloud Router	BGP control plane for dynamic route exchange
External Peer VPN Gateway	Represents the FortiGate and its two public ISP endpoints
FortiGate	On-premises VPN endpoint and routing/security device
ISP1	Primary VPN transport path
ISP2	Redundant VPN transport path
IPsec Tunnel 1	VPN path associated with ISP1
IPsec Tunnel 2	VPN path associated with ISP2
BGP	Dynamic routing between GCP and FortiGate

Setup Steps

The implementation was completed through the following high-level stages.

1. GCP Configuration

1. Created the required **GCP VPC** and networking components.
2. Created a **Cloud HA VPN gateway**.
3. Created a **Cloud Router** with dynamic BGP configuration.
4. Created an **external peer VPN gateway** representing the FortiGate.
5. Configured:
 - Interface 0 → FortiGate ISP1 public IP
 - Interface 1 → FortiGate ISP2 public IP
6. Created the two HA VPN tunnels.
7. Configured the corresponding BGP interfaces and peers.
8. Used the automatically assigned GCP BGP link-local addressing for the tunnel/BGP relationships.

2. FortiGate IPsec Configuration

Two custom route-based IPsec tunnels were created on the FortiGate.

Each tunnel is bound to a different physical WAN interface:

```
Tunnel 1 → ISP1 / WAN1
Tunnel 2 → ISP2 / WAN2
```

The IPsec configuration included:

- Matching Phase 1 parameters
- Matching pre-shared keys
- Matching Phase 2 parameters
- Route-based tunnel configuration
- 0.0.0.0/0 Phase 2 selectors
- Tunnel interface addressing using the GCP-assigned 169.254.X.X/30 ranges
- Appropriate local and remote tunnel IP configuration

The two tunnels provide independent transport paths to the GCP HA VPN gateway.

3. FortiGate BGP Configuration

BGP was configured on the FortiGate using the ASN provided for the GCP Cloud Router relationship.

Two BGP neighbors were configured:

```
BGP Neighbor 1 → GCP Cloud Router Peer 1 → ISP1 tunnel
BGP Neighbor 2 → GCP Cloud Router Peer 2 → ISP2 tunnel
```

The FortiGate advertises the required on-premises LAN network through a **scoped network/route-map configuration**.

The implementation intentionally avoids blanket **redistribute-connected** behavior so that unrelated connected networks—such as WAN, transit, or infrastructure subnets—are not unintentionally advertised toward GCP.

4. Firewall Policies

Firewall policies were configured to permit the required traffic between the on-premises LAN and the GCP networks across the VPN tunnels.

The policies were reviewed to ensure:

- Required LAN-to-GCP traffic is permitted.
- Required GCP-to-LAN traffic is permitted.
- The IPsec interfaces are correctly referenced.
- Routing and firewall policy behavior are consistent with the intended traffic flow.
- Security inspection and logging requirements are applied according to the existing FortiGate security policy.

Verification & Testing

The completed implementation was validated from both the VPN and routing perspectives.

IPsec Validation

- ☐ IPsec Tunnel 1 established successfully.
- ☐ IPsec Tunnel 2 established successfully.
- ☐ Both tunnels use the intended ISP/WAN interfaces.
- ☐ Phase 1 negotiation completed successfully.
- ☐ Phase 2 negotiation completed successfully.
- ☐ Route-based tunnel interfaces are operational.

BGP Validation

- ☐ BGP Neighbor 1 reached **Established** state.
- ☐ BGP Neighbor 2 reached **Established** state.
- ☐ Both BGP sessions remain operational while both ISP paths are available.
- ☐ GCP and FortiGate exchanged the expected routes.
- ☐ On-premises LAN prefixes were advertised toward GCP.
- ☐ GCP prefixes were learned correctly by the FortiGate.
- ☐ Unintended WAN/transit prefixes were not advertised.

Connectivity Validation

- ☐ On-premises LAN-to-GCP connectivity tested successfully.
- ☐ GCP-to-on-premises LAN route exchange validated.
- ☐ Traffic successfully traversed the IPsec VPN paths.
- ☐ Routing behavior was verified using the learned BGP routes.

HA / Failover Validation

- ☐ ISP1 VPN path tested.
- ☐ ISP2 VPN path tested.
- ☐ Failover behavior between the two ISP paths tested successfully.
- ☐ VPN connectivity remained available during the tested ISP path transition.
- ☐ BGP behavior was validated during the failover testing.
- ☐ Recovery/reversion behavior was tested after the primary path became available again.

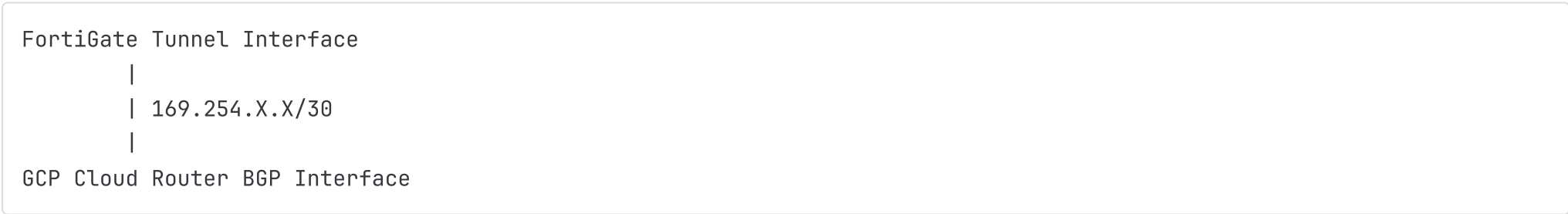
Lessons Learned / Gotchas

1. Route-Based IPsec Tunnel IPs Must Be Configured Explicitly

A route-based IPsec tunnel interface does not automatically become a usable routed interface simply because the IPsec tunnel is established.

For BGP-based VPN routing, the tunnel interfaces need explicit local and remote addressing corresponding to the GCP-assigned link-local network.

Conceptually:



The tunnel being **UP** and the tunnel interface being correctly configured for routing are separate considerations.

2. Use 0.0.0.0/0 Phase 2 Selectors for BGP-Routed VPNs

For this design, Phase 2 selectors use:

Local Selector : 0.0.0.0/0

Remote Selector : 0.0.0.0/0

This is important because the actual networks exchanged through the VPN are determined by the routing control plane rather than by narrowly defined Phase 2 subnet selectors.

3. Scope BGP Route Advertisement

Avoid using blanket connected-route redistribution unless every connected network is intentionally supposed to be advertised.

For example, the FortiGate may have connected networks representing:

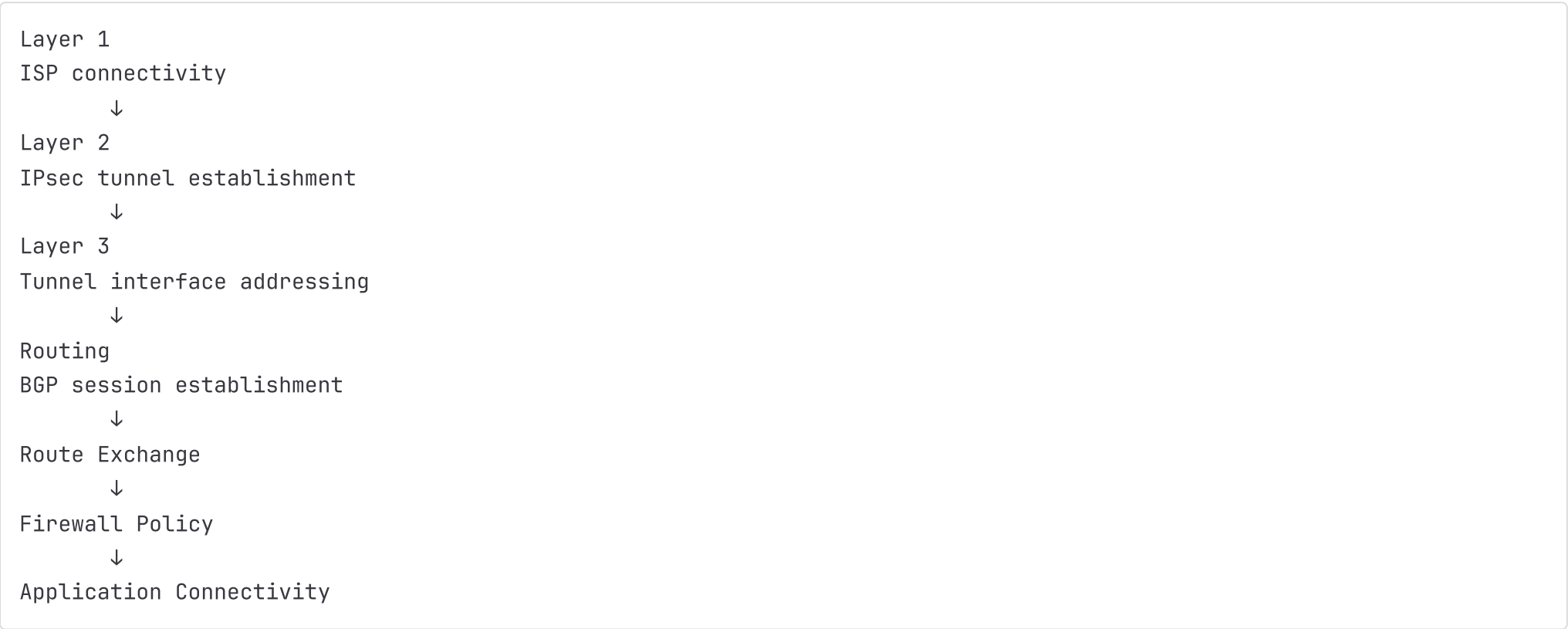
```
LAN
WAN / ISP
Transit
Management
VPN infrastructure
Other internal interfaces
```

Only the required prefixes should be advertised toward GCP.

A scoped network/route-map approach provides better control and prevents accidental leakage of WAN, transit, or infrastructure routes into the cloud routing domain.

4. Separate the VPN Transport From the Routing Control Plane

The design becomes easier to troubleshoot when the layers are considered independently:



A tunnel being established does not necessarily mean BGP is established, and an established BGP session does not necessarily mean the required firewall policy or application connectivity is correct.

Production Outcome

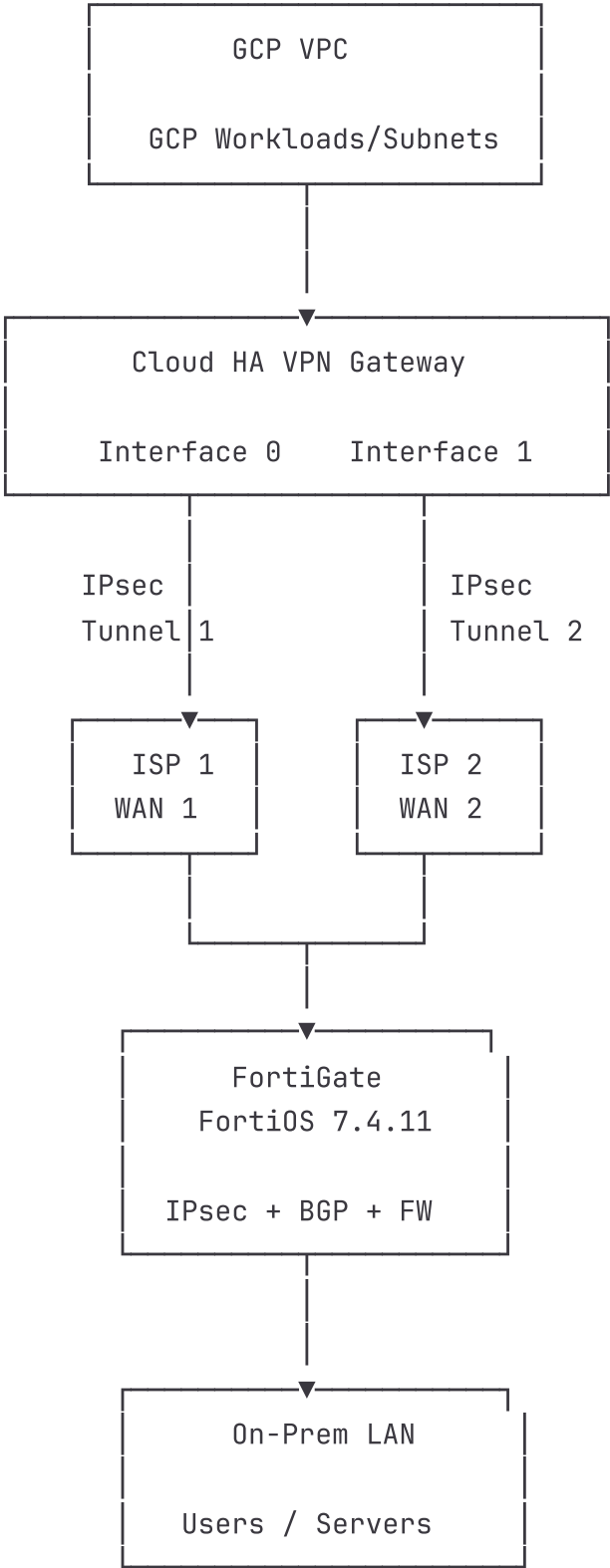
The GCP Cloud HA VPN ↔ FortiGate integration is **fully operational in production**.

The completed design provides:

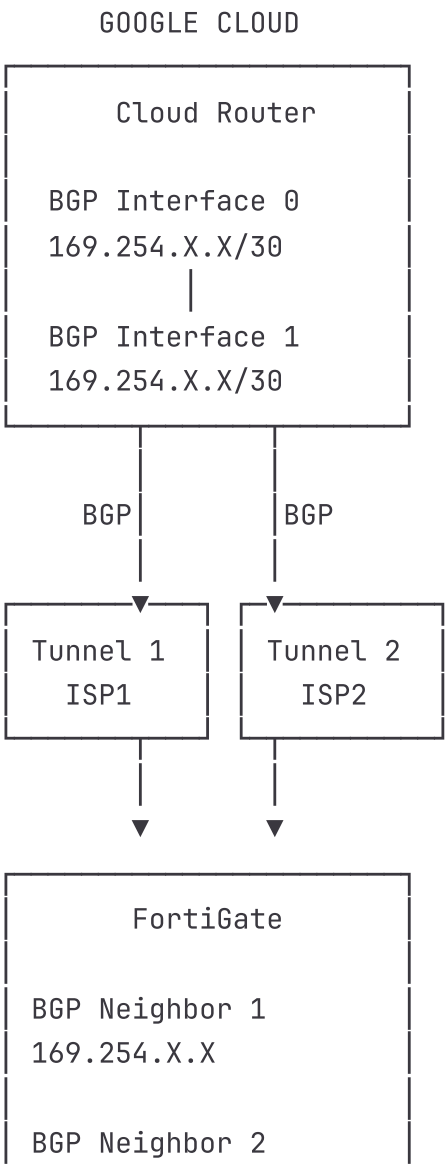
- Dual ISP VPN connectivity
- Two independent route-based IPsec tunnels
- Dynamic BGP routing
- Bidirectional route exchange
- Redundant GCP-to-on-premises connectivity
- Controlled LAN route advertisement
- Tested ISP failover behavior

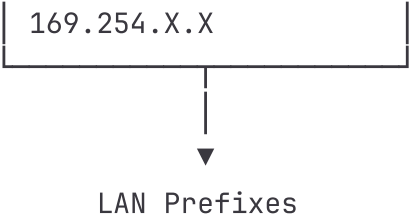
The final architecture provides a maintainable foundation for resilient connectivity between the on-premises environment and GCP while avoiding dependence on manually maintained static VPN routes.

Production Architecture / Traffic Flow

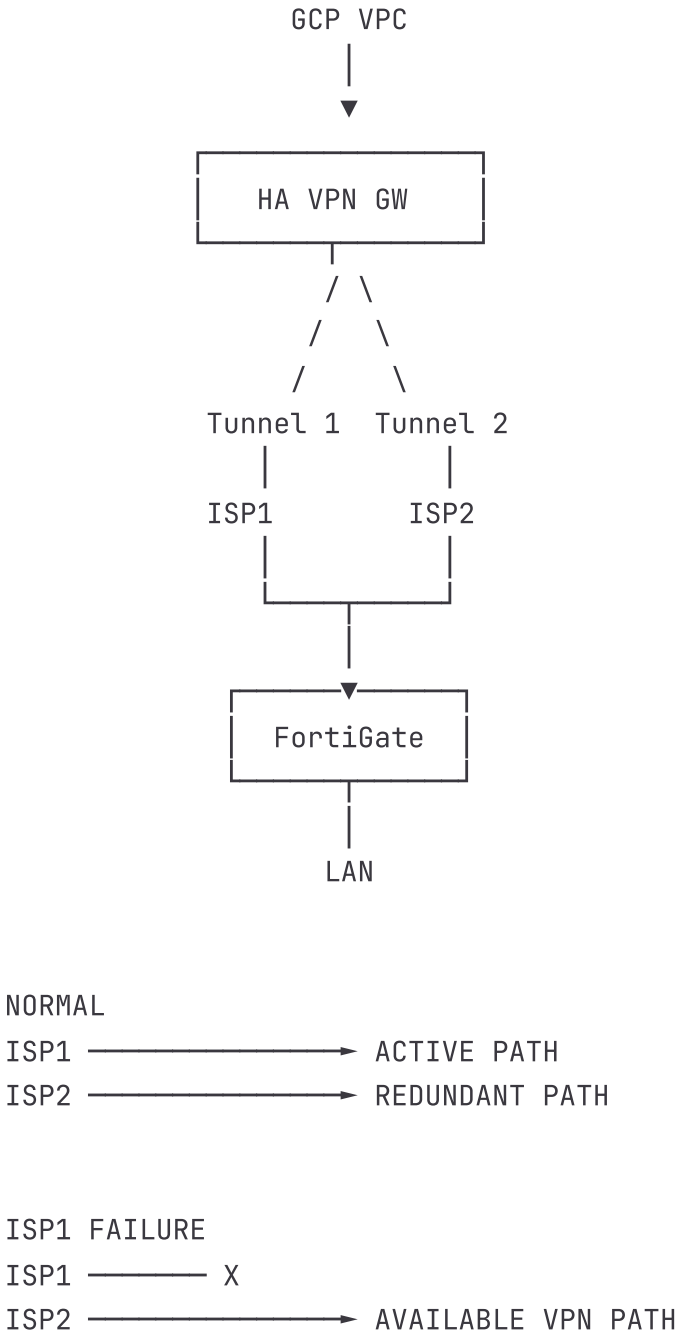


BGP Control-Plane Diagram





HA Diagram



Comments